



CODEBOOK

Chilean Political Landscape Dataset

CPLD • Version 4.0 • 1989-2025

FILE	cpld_v4_0 [.tab/.txt/.csv/.xlsx]
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VERSION	4.0
OBSERVATIONS	6,119
VARIABLES	46
COVERAGE	1989-2025
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WEBSITE	labdemgob.github.io/cpld
LICENSE	CC BY 4.0

HOW TO READ THIS CODEBOOK

Each entry names a variable, gives its definition, and where applicable the formula used to compute it, its type and unit, and its full source. Variables are numbered sequentially from the identification block onward. Entries sourced from the Servicio Electoral de Chile (SERVEL) are the author’s elaboration based on certified electoral results.

Unit of observation. Each row represents one electoral unit in one election. The electoral unit varies by election type: **district** for Diputados and Constituyentes, **region** for Senadores and Gobernadores, **province** for Cores, and **commune** for Alcaldes and Concejales. Row counts therefore equal unit counts: Diputados 2021 has 28 rows (28 districts); Alcaldes 2024 has 345 rows (345 communes).

ANATOMY OF AN ENTRY

Number · name · type	Running variable number, the column name, and its data type or unit.
Definition	What the variable measures and how it is constructed.
Formula	The computation, shown in a tinted box where one applies.
Source	Full APA reference, repeated in each entry for self-contained citation.

NOTATION USED IN FORMULAS

v_i – vote share of list i	s_i – seat share of list i
p_i – unit vote proportion	P_i – national vote proportion
w_i – weight (vote share)	x_i – ideological score of list i
dm – district magnitude	n – number of lists in the unit

Warning – unit is not unique. Unit identifiers are not unique across electoral systems, because numbering restarted when districts were redrawn. Ids 1-28 exist in both the pre- and post-2017 deputy schemes and refer to different territories. Group or join on **type2** together with unit and election.

01 IDENTIFICATION VARIABLES

01 **type** STRING

Type of election. Values: Diputados, Senadores, Constituyentes, Alcaldes, Concejales, Cores, Gobernadores, Municipales (elections prior to the split of Alcaldes/Concejales).

Source – Author.

02 **type2** STRING

Electoral system group. Identifies sets of elections that share the same electoral rules (district boundaries, district magnitudes, seat totals). Examples: Diputados 1989–2013 (60 districts, binominal); Diputados 2017–2021 (28 districts, proportional); Senadores Binominal (Pares/Impares); Alcaldes 2004–2021; Cores 2024.

Source – Author. Electoral system classification follows: Lijphart, A. (1994). *Electoral Systems and Party Systems*. Oxford University Press.

03 **type3** STRING

Election label. Concatenation of type and election year, e.g. “Diputados - 2021”. type3 is a label, not a unique row key: Gobernadores first and second rounds share the same type and year, so “Gobernadores - 2021” and “Gobernadores - 2024” each appear twice, 48 rows in total. Only type2 separates them.

Source – Author.

04 **region_2024** NUMERIC

Regional identifier using 2024 regional boundaries, enabling longitudinal comparison across elections despite historical boundary changes. Notable assignments: Diputados district 42 → region 16; Senadores circ. 1 (Arica y Tarapacá) → region 15; Senadores circ. 13 (Bio-Bio Cordillera) → region 8.

Source – Author.

05 **unit** INTEGER / STRING

Identifier of the electoral unit. Integer for district- and region-based elections; string for commune-based elections. Not unique across electoral systems – see the warning on page 2.

Source – Author with SERVEL.

06 **election** INTEGER · 1989–2025

Year of the election.

Source – Author with SERVEL.

02 LIST STRUCTURE VARIABLES

07 **list** INTEGER

Number of electoral lists competing in the unit. Counts all distinct lists including IFP candidates; each IFP candidate constitutes one list.

Source – Author with SERVEL.

08 **lists_national** INTEGER

Number of distinct electoral lists competing nationally in the election, excluding IFP candidates. Measures the national menu of partisan options.

Source – Author with SERVEL.

09 **listmin** NUMERIC · %

Minimum vote share (%) obtained by any list in the unit.

Source – Author with SERVEL.

10 **listmax** NUMERIC · %

Maximum vote share (%) obtained by any list in the unit.

Source – Author with SERVEL.

11 **candidates** INTEGER

Total number of individual candidates competing in the unit, excluding null and blank votes.

Source – Author with SERVEL.

12 **ifp_candidates** INTEGER

Number of Independientes Fuera de Pacto (IFP) candidates competing in the unit – independents running outside any coalition or party list.

Source – Author with SERVEL.

13 **ifp_seats** INTEGER

Number of seats won by IFP candidates in the unit.

Source – Author with SERVEL.

14 **percent_ind** NUMERIC · %

Percentage of total candidates who are IFP candidates.

```
percent_ind = (ifp_candidates / candidates) × 100
```

Source – Author.

03 ELECTORAL SYSTEM VARIABLES

15 **dm** INTEGER

District magnitude – number of seats allocated in the unit. For majoritarian elections (Alcaldes, Gobernadores), $dm = 1$; for proportional elections, dm varies by unit.

Source – Author with SERVEL electoral rules.

16 **as** INTEGER

Assembly size. The total number of seats in the assembly to which the unit's seats belong: for Alcaldes/Gobernadores, $as = dm = 1$ (a single office); for Concejales/Municipales, $as = dm$ (the concejo is constituted at the comuna level); for Cores, as is the full regional council size, the sum of dm across all provincias of the region; for Diputados, Senadores and Constituyentes, $as = as2$, the nationally contested total. Because as is defined at the level each assembly is constituted, the seat product ($sp = dm \times as$) is comparable within an election type but not across types – a pooled model should use $as2$.

Source – Author.

17 **as2** INTEGER

Assembly size (national). Total seats allocated nationally in the election, the sum of dm across all units of the election. Unlike as , $as2$ is defined identically for every election type, so it is the term to use when pooling across types. For Diputados, Senadores and Constituyentes, $as2 = as$.

Source – Author.

18 **sp** NUMERIC

Seat product – key predictor of party system fragmentation in the Seat Product Model. Comparable within an election type but not across types (see as).

$$sp = dm \times as$$

Source – Author with: Shugart, M. S., & Taagepera, R. (2017). *Votes from seats: Logical models of electoral systems*. Cambridge University Press. <https://doi.org/10.1017/9781316342787>

19 **spm** NUMERIC

Seat Product Model predicted ENP – theoretical prediction of the effective number of parties from the seat product.

$$spm = (dm \times as)^{(1/6)} = sp^{(1/6)}$$

Source – Author with Shugart & Taagepera (2017). See sp .

20 **logmag** NUMERIC

Natural logarithm of district magnitude.

$$\text{logmag} = \ln(\text{dm})$$

Source — Author.

04 PARTY SYSTEM FRAGMENTATION

21 **enpv** NUMERIC

Effective number of parties by votes at the unit level. IFP candidates are included. Caution: this column is not the unit-level input to Chhibber_Kollman or cox_inflation, which recompute unit ENPv with IFPs excluded, matching enpv_national.

$$\text{ENPv} = 1 / \sum p_i^2$$

Source — Laakso, M., & Taagepera, R. (1979). "Effective" number of parties: A measure with application to West Europe. *Comparative Political Studies*, 12(1), 3-27. <https://doi.org/10.1177/001041407901200101>

22 **enps** NUMERIC

Effective number of parties by seats at the unit level. IFP candidates included.

$$\text{ENPs} = 1 / \sum s_i^2$$

Source — Laakso & Taagepera (1979). See enpv.

23 **enpv_national** NUMERIC

Effective number of parties by votes at the national level, computed on aggregated national vote totals, excluding IFP candidates.

$$\text{ENPv_national} = 1 / \sum P_i^2$$

Source — Laakso & Taagepera (1979). See enpv.

24 **enps_national** NUMERIC

Effective number of parties by seats at the national level, excluding IFP candidates.

$$\text{ENPs_national} = 1 / \sum S_i^2$$

Source — Laakso & Taagepera (1979). See enpv.

25 **dif** NUMERIC

Difference between the raw number of lists and ENPv at the unit level — how many "effective" lists are lost to fragmentation of small parties.

$$\text{dif} = \text{list} - \text{enpv}$$

Source — Author.

26 **dif2** NUMERIC

Difference between ENPv and ENPs at the unit level, reduction in effective parties from translating votes into seats. The floor at zero is substantive, not cosmetic: in 97 rows enps exceeds enpv, so the unfloored difference is negative and dif2 is stored as zero. Integer seat allocation can distribute seats more evenly than votes were distributed. Users who need the signed quantity should compute $\text{enpv} - \text{enps}$ directly.

$$\text{dif2} = \max(0, \text{enpv} - \text{enps})$$

Source – Author.

05 DISPROPORTIONALITY INDICES

Notation: v_i = vote share of list i (proportion); s_i = seat share of list i (proportion); n = number of lists.

27 **Rae**

Rae disproportionality index.

$$\text{Rae} = (1/n) \cdot \sum |v_i - s_i|$$

Source – Rae, D. W. (1967). *The Political Consequences of Electoral Laws*. Yale University Press.

28 **LH**

Loosemore-Hanby disproportionality index.

$$\text{LH} = 0.5 \cdot \sum |v_i - s_i|$$

Source – Loosemore, J., & Hanby, V. J. (1971). The theoretical limits of maximum distortion. *British Journal of Political Science*, 1(4), 467-477. <https://doi.org/10.1017/S000712340000925X>

29 **Lijphart_1**

Largest deviation from proportionality among lists holding at least one seat.

$$\text{Lijphart}_1 = \max |v_i - s_i|, \text{ for } s_i > 0$$

Source – Lijphart, A. (1990). The political consequences of electoral laws, 1945-1985. *American Political Science Review*, 84(2), 481-496. <https://doi.org/10.2307/1963530>

30 **Lijphart_2**

Largest deviation from proportionality among all lists.

$$\text{Lijphart}_2 = \max |v_i - s_i|, \text{ for all } i$$

Source – Lijphart, A. (1994). *Electoral Systems and Party Systems*. Oxford University Press.

31 Gallagher NUMERIC

Gallagher least-squares disproportionality index over the lists of the unit. Range: 0 to about 0.9, bounded above by LH on every row – a theorem, now enforced in the script's validation block.

$$G = \sqrt{0.5 \cdot \sum (v_i - s_i)^2}$$

Source – Author with: Gallagher, M. (1991). Proportionality, disproportionality and electoral systems. *Electoral Studies*, 10(1), 33-51. [https://doi.org/10.1016/0261-3794\(91\)90004-C](https://doi.org/10.1016/0261-3794(91)90004-C)

32 Cox_Shugart NUMERIC

Regression-based measure of the relationship between vote and seat shares. A value of 1.0 indicates perfect proportionality; values above 1.0 indicate a bonus to larger parties. Chilean district values sit above 1 almost everywhere, as they should under d'Hondt.

$$\text{Cox_Shugart} = b, \text{ where } s_i = a + b \cdot v_i \text{ (OLS)}$$

Note: missing where fewer than three lists contested the unit; see Missing values, Section 10.

Source – Author with: Cox, G. W., & Shugart, M. S. (1991). Comment on Gallagher's "Proportionality, disproportionality and electoral systems." *Electoral Studies*, 10(4), 348-352. [https://doi.org/10.1016/0261-3794\(91\)90025-N](https://doi.org/10.1016/0261-3794(91)90025-N)

06 PARTY SYSTEM NATIONALIZATION

All six variables in this section are computed directly from the vote data. Three conventions apply throughout: IFP lists are excluded; units a list did not contest count as zero; the electorate of a unit is its valid vote to non-IFP lists.

33 Mainwaring_Jones NUMERIC · 0-100

Party system nationalization score, the vote-weighted mean of each list's nationalization score, where a list's score is one minus the Gini coefficient of its vote distribution across the units of the election. Higher means support is spread more evenly.

$$\text{PNS}_i = 1 - \text{Gini}(p_{i1}, \dots, p_{id}) \quad \cdot \quad \text{PSNS} = 100 \cdot \sum (s_i \cdot \text{PNS}_i)$$

Caution: not comparable across election types – Gini-based scores fall mechanically as units rise (7 to 345 in this dataset). A controlled resample of Alcaldes 2021 puts the artefact at 50.5 points merging 343 units to 4. Use `pns_standardised` for cross-type comparisons.

Source – Author with: Jones, M. P., & Mainwaring, S. (2003). The nationalization of parties and party systems. *Party Politics*, 9(2), 139-166. <https://doi.org/10.1177/13540688030092002>

34 **pns_weighted** NUMERIC · 0-100

Party nationalization score with weighted units. As Mainwaring_Jones, but the Gini is taken over the cumulative distribution of voters rather than of units, so units count in proportion to their electorates. Returns exactly 100 under perfect homogeneity.

$$\text{PNSw} = 2 \cdot \sum (v_i \cdot (\sum_{j \leq i} p_j - p_i/2)) / (\sum v_i \cdot \sum p_i), \text{ vote-weighted mean across lists} \times 100$$

Caution: weighting by electorate does not correct for the number of units – the same Alcaldes 2021 resample moves this variable 46.3 points. Not comparable across election types.

Source – Author with: Bochsler, D. (2010). Measuring party nationalisation. *Electoral Studies*, 29(1), 155-168, eq. p.162. <https://doi.org/10.1016/j.electstud.2009.06.003>

35 **pns_standardised** NUMERIC · 0-100

pns_weighted corrected for the number of territorial units, standardising every election to a ten-unit baseline. The most comparable of the three nationalization measures across election types, though not fully comparable.

$$E = (\sum v_i)^2 / \sum v_i^2 \quad \cdot \quad \text{sc-camel-s-p-n-s} = \text{PNSw} ^ (1/\log_{10} E)$$

Note: the same Alcaldes 2021 resample moves this variable 20.7 points, about three fifths of the unit artefact removed. Anchored at ten units; senate cycles (effective units ≈4) rest on extrapolation. See note 19 of the cautionary notes.

Source – Author with Bochsler (2010), eq. p.164.

36 **Chhibber_Kollman** NUMERIC

Indicator of party aggregation: the effective number of parties nationally minus the mean effective number of parties across units, both on votes. Zero means national fragmentation matches the average local one.

$$\text{Chhibber_Kollman} = \text{ENPv_national} - (1/d) \cdot \sum \text{ENPv_j}$$

Caution: the ENPv_j here are recomputed with IFP lists excluded, matching enpv_national – NOT the published enpv column. The two bases diverge most where independents are numerous (Alcaldes 2024: 3.02 vs 1.89).

Source – Author with: Chhibber, P., & Kollman, K. (1998). Party aggregation and the number of parties in India and the United States. *American Political Science Review*, 92(2), 329-342. <https://doi.org/10.2307/2585667>

37 **cox_inflation** NUMERIC · 0-100

Chhibber_Kollman expressed as a share of the national effective number of parties – the percentage of national fragmentation from aggregating across units rather than within-unit competition. Alcaldes 2021 scores 56.7.

$$\text{cox_inflation} = 100 \cdot (\text{ENPv_national} - \text{mean}(\text{ENPv_j})) / \text{ENPv_national}$$

Note: can go slightly negative (Constituyentes 2023: -0.20; Senadores 1989: -0.94), a real result. Allik's (2006) index equals 100 minus this value and is not carried separately.

Source – Author with: Cox, G. W. (1999). Electoral rules and electoral coordination. *Annual Review of Political Science*, 2, 145-161. <https://doi.org/10.1146/annurev.polisci.2.1.145>

38 territorial_coverage NUMERIC · 0-100

Percentage of list-by-unit combinations actually contested, ignoring vote shares. Alcaldes 2021 scores 21.4 against 76.7 for Municipales 1992. IFP lists excluded.

$$T = 100 \cdot (\text{list-unit pairs contested}) / (\text{lists} \times \text{units})$$

Source – Author with: Caramani, D. (2004). *The Nationalization of Politics*. Cambridge University Press, p.61.
<https://doi.org/10.1017/CB09780511616662>

07 IDEOLOGICAL VARIABLES**39 av_id** NUMERIC · 1-10

Vote-weighted mean ideological score of the district, excluding IFP (score = 5). Where the match is partial, weights are renormalised within the matched subset. Scale: 1 (far left) to 10 (far right).

$$\text{av_id} = \sum(w_i \cdot x_i) / \sum w_i$$

Source – Bunker, K. (2025). Mapping party systems over time. *Journal of Political Ideologies*.
<https://doi.org/10.1080/13569317.2025.2452179>

40 polarization_district NUMERIC

Ideological polarization at the unit level, vote-weighted dispersion across the spectrum, excluding IFP. Missing where every competing list was an IFP (see cautionary notes).

$$P = \sqrt{(\sum w_i \cdot (x_i - \bar{x})^2) / \sum w_i}$$

Source – Dalton, R. J. (2008). The quantity and quality of party systems. *Comparative Political Studies*, 41(7), 899-920. <https://doi.org/10.1177/0010414008315860>

41 polarization_national NUMERIC

Same formula as polarization_district, applied to national vote shares, excluding IFP.

Source – Dalton (2008). See polarization_district.

08 CONTEXTUAL VARIABLES**42 ethnic_frac** NUMERIC · [0,1]

Ethnic fractionalization – probability two randomly selected individuals belong to different ethnic groups (2017 census). 0 = homogeneous, 1 = maximally fractionalized.

$$F = 1 - \sum (n_g / N)^2$$

Source – Alesina, A., et al. (2003). Fractionalization. *Journal of Economic Growth*, 8(2), 155-194.
<https://doi.org/10.1023/A:1024471506938>

43 ethnic_presence NUMERIC · [0,1]

Share of the population identifying as indigenous (2017 census).

```
ethnic_presence = n_indigenas17 / censo17
```

Source – Author with Censo (2017).

44 ethnic_number NUMERIC · %

Share of the population identifying as indigenous (2002 census). Missing for units created after 2004, see Section 10.

```
ethnic_number = (n_indigenas02 / censo02) × 100
```

Source – Author with Censo (2002).

09 VOTE AND STRUCTURAL VARIABLES

45 votes INTEGER

Total number of valid votes cast in the unit. Used as a proxy for district size and urban-rural standing.

Source – Author with SERVEL certified results.

46 subunits INTEGER

Number of comunas composing the electoral unit. For commune-level elections (Alcaldes, Concejaes), subunits = 1; for district or regional elections, reflects how many comunas fall within the unit boundaries.

Source – Author.

10 MISSING VALUES

Blank cells in this dataset are not interchangeable. Each pattern below has a distinct cause, and users should decide case by case whether to drop, impute, or treat the row as informative. Patterns are listed by the variable affected.

Cox_Shugart

Missing on 492 rows, the units contested by fewer than three lists. Nine had a single list, where vote share has no variance and the slope is undefined outright. The other 483 had two lists, where a slope through two points is the difference ratio $(s1-s2)/(v1-v2)$, reaching 4,091 in Zapallar 2016. Those rows are set to missing deliberately, not released as numbers that would dominate any model fitted on the column. LH and Gallagher already carry the information.

This is a deliberate restriction, not a gap in the source data. Where three or more lists ran and every one of them won an identical number of seats, the slope is a genuine zero rather than a blank. Eight such rows.

polarization_district

Missing in comunas where every competing list was an Independiente Fuera de Pacto. Polarization is computed over lists carrying an ideological score, and IFPs are excluded by construction (see cautionary notes), so a comuna contested only by IFPs leaves nothing to compute over. Affects a small number of Alcaldes comunas in 2016, 2021 and 2024.

av_id

Missing wherever no party competing in the unit can be matched to the ideological scores input.

Note on construction, which applies whether or not any value is missing. Where the match is partial rather than empty, av_id is a vote-weighted average over the matched parties only, with weights renormalised within that subset rather than over all votes cast in the unit.

ethnic_number

Missing for Alto Biobío, Alto Hospicio, Cholchol and Hualpén in every municipal election. All four comunas were created in 2004, after the 2002 census that supplies the indigenous population and total population counts behind this variable. ethnic_frac is populated for these same rows because it draws on a different source, so the two ethnic measures do not share the same coverage.

ACCESS THE DATASET



SCAN FOR THE PROJECT PAGE

labdemgob.github.io/cpld

Downloads, codebook, version history, and replication materials for the CPLD.

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